



Hidden Radiology

Ana Fahim Series

Radiation Safety

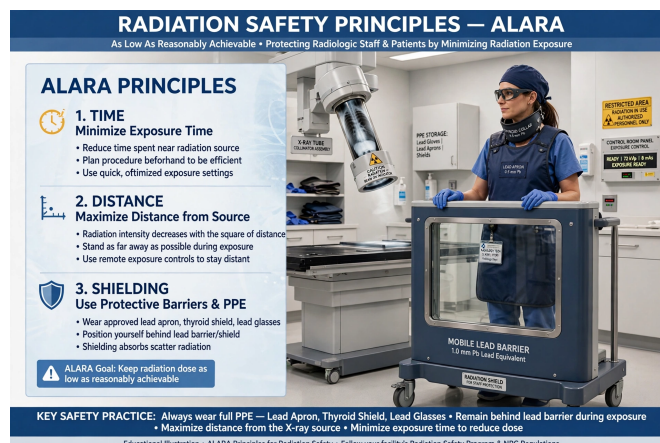
Aman El-Ish3a3 - Ana Fahem

Premium Edition | Ready for Gumroad, Payhip & Sellfy | Hidden Radiology 2026

Dr. Nasser Mabrouk | radiologyhidden | Commercial License Included

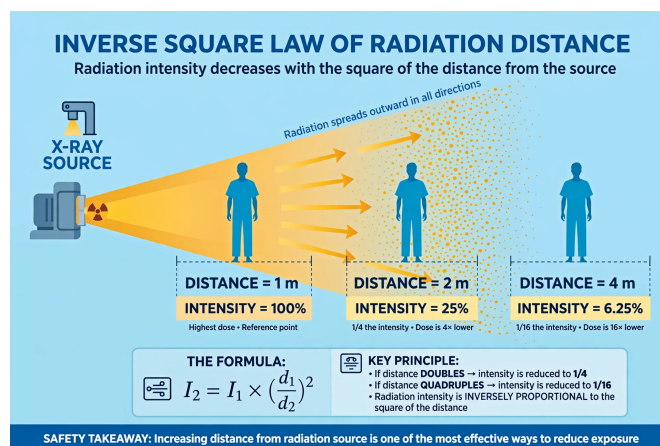
1. WHY RADIATION SAFETY MATTERS / Limaza Aman al-Ish3a3 Mohem

- ALARA: As Low As Reasonably Achievable - The Golden Rule of Radiology.
- Three Pillars: TIME (reduce time) | DISTANCE (increase distance) | SHIELDING (use protection).
- Justification: Every exposure must have benefit > risk. No routine, no habit.
- Optimization: Best diagnostic quality with lowest dose. Not lowest dose only.
- Scatter radiation is the main staff hazard in fluoroscopy and mobile X-ray.
- Key Insight (Arabic): Al-ish3a3 ghayr mar'i laken atharuhu haqiqi. Kull thania tafreq.



2. DISTANCE - Your Invisible Shield

- Inverse Square Law: $I_2 = I_1 \times (d_1/d_2)^2$
- At 1m: 100% dose | At 2m: 25% dose (75% reduction) | At 4m: 6.25% (93.75% reduction)
- Practical rule: Step back 2 meters during mobile chest X-ray = 4x protection.
- Never hold cassette with hands in primary beam - use holder.
- Positioning: Use longest possible tube-to-patient distance.
- Arabic Summary: Law of distance: If you double distance, dose drops to quarter.



3. SHIELDING & PPE - Your Armor

- Lead Apron 0.5mm Pb: Attenuates ~90% scatter at 75kVp, ~99% if not in primary beam.
- Types: Front-only (0.5) vs Wrap-around (0.25 front+0.25 back = 0.5 front). Wrap better.
- Thyroid Shield 0.5mm Pb: Thyroid dose limit critical, cancer risk high.
- Lead Glasses 0.5mm Pb: Lens cataract threshold 0.5 Gy acute, 5 Gy chronic. Limit 20 mSv/yr.
- Dosimeter: Wear at collar outside apron, facing tube. For pregnant: second badge under apron at waist.
- QA: Inspect aprons annually under fluoro for cracks. Fold never, hang always.



4. PATIENT SAFETY - Collimation & Gonads

- Collimation: Reduces field to anatomy of interest. Reduces scatter 40-70% + improves contrast.
- Gonadal Shield: Use if gonads within 5cm of field AND not obscuring anatomy. 0.5mm Pb.
- Contact shield vs Shadow shield: Shadow mounted on tube better, no misplacement.
- Female patients: Always ask LMP, document. Use lead apron for pelvis if outside field.
- Dose tracking: Record DAP / CTDIvol / DLP in report. Compare to DRLs.

5. SPECIAL POPULATIONS - Pediatrics & Pregnancy

- Pediatrics: 10x radiosensitivity, longer life expectancy = higher risk.
- Pediatric techniques: Lower kVp/mAs, no anti-scatter grid for <12cm thickness, fast exposure, immobilization devices.
- Image Gently: One shot, good quality, no repeats. Use paediatric protocols.
- Pregnancy Patient: If exam justified and cannot wait, proceed with abdominal shielding 0.5mm Pb + informed consent.
- Fetal dose <50 mGy: No increased risk of malformation. <100 mGy: No termination justification per ICRP.
- Pregnant Staff: After declaration, fetal limit 1 mSv remainder of pregnancy. Can continue work with extra badge.

6. DOSE LIMITS, MONITORING & SAFETY CULTURE

- Occupational (ICRP 103): 20 mSv/yr averaged over 5 years (max 50 in one year).
- Lens of eye: 20 mSv/yr | Skin, hands, feet: 500 mSv/yr | Public: 1 mSv/yr.
- Pregnant worker fetus: 1 mSv after declaration.
- Dosimetry: Monthly reading, investigate if >0.5 mSv/month unexpectedly.
- Safety Culture Checklist BEFORE exposure:
 - - Justified? Is there non-ionizing alternative (US/MRI)?
 - - Pregnant? LMP checked and documented?
 - - Collimated tightly?
 - - Shielded patient and staff?
 - - ALARA settings applied?
- Final Note: Culture eats checklist for breakfast. Teach, audit, report near-miss.

BONUS - Ready to Sell Pack

Included in this Premium Pack for Gumroad:

- [1] Premium PDF (this file) - 300 DPI, print-ready
- [2] 24 High-Res Illustrations from Quality-Safety Library
- [3] Editable Canva Cover Template Link
- [4] ALARA Poster A3 - PNG + PDF
- [5] Radiation Safety Checklist - Printable
- [6] Pregnancy Screening Form - Arabic/English
- [7] Commercial License (sell to clinics, teach, bundle)
- [8] Product Description + Keywords for Gumroad SEO

Suggested Pricing: \$19 (basic) - \$49 (pro bundle)

License: You can resell, you cannot share source for free

Brand: Hidden Radiology / radiologyhidden